Separating decision rule failures from representation failures in external validation of diabetic retinopathy grading

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Abstract

Diabetic retinopathy grading models lose performance on new populations, and the losses are usually attributed to the learned representation. In a pre-registered study of ten development models evaluated on APTOS and DDR, five seed replicates and a reverse-direction model, we separated that contribution from the ordinal decision rule that converts scores into grades. Discrimination of referable disease transferred (area under the curve 0.914–0.973), but the models over-graded APTOS and under-graded DDR: a threshold fitted for 85% development sensitivity reached 99.33% and 63.38%. Refitting cut-points alone, without retraining, restored APTOS mild-grade recall from 1.6–14.3% to 61.6–77.8% and raised reverse-direction referral sensitivity from 43.5% to 77.9%. Contrary to our pre-registered predictions, model choice mattered more than development-fitted rules, and the mild-grade collapse did not generalise. Quadratic weighted kappa varied by under 0.01 across DDR operating points spanning 71.3–86.3% referral sensitivity. External evaluations should report operating curves; deployment should include local recalibration.

Keywords: diabetic retinopathy, ordinal grading, decision threshold, calibration, external validation, foundation models, diabetic macular oedema

Introduction

Diabetic retinopathy (DR) affects roughly one in five people with diabetes, an estimated 103 million adults in 2020 and a projected 161 million by 2045, and it remains

a leading cause of preventable blindness in the working-age population[1]. Diabetic macular oedema (DME), its most common mechanism of central vision loss, affects about one in fifteen[2]. Deep learning systems that grade fundus photographs reached specialist-level agreement in development studies a decade ago[3,4], and autonomous screening devices have since been authorised on the strength of prospective pivotal trials[5,6]. Yet the same class of system behaves very differently once it leaves the population it was built on. In a head-to-head evaluation of seven commercial algorithms on the same real-world images, sensitivity for referable disease ranged from 51% to 86%[7]; three commercial algorithms applied to the same eyes against the same reference standard in Indian public-health clinics returned specificities between 14% and 96%[8].

When a model fails on a new population, the failure is usually read as a statement about what the network has learned. A recent frozen-encoder benchmark found that two of three encoders, including the retinal foundation model RETFound, did not identify a single one of 270 mild cases in an external corpus (grade-1 F1 = 0.000), and attributed the collapse to the encoders' representations[9]. RETFound itself was shown to be superior using macro-averaged, threshold-free metrics[10], which cannot reveal a per-grade failure of this kind. Comparisons between architectures, heads and pretraining schemes are reported in the same way: as differences in what a model represents.

An ordinal grading model does not output a grade, however. It outputs scores that a decision rule turns into a grade, and the rule is almost never examined. Three different things are routinely conflated: *probability calibration*, whether predicted probabilities match observed frequencies[11,12]; the *decision rule*, the cut-points or referral threshold that map scores to a grade or a referral; and the *representation*, what the encoder has learned. The common default, a threshold of 0.5 on every ordinal output or an argmax, is a hyper-parameter that has been chosen implicitly rather than one that has been avoided. If a model ranks images correctly but its cut-points sit in the wrong place for a new population, it will look blind to a grade it in fact separates, and two models that differ mainly in where their defaults happen to fall will look as if they differ in what they learned. Clinical prediction research has long treated recalibration and model updating as a routine step before deployment[13,14]; DR grading studies rarely test whether their headline failures and gains survive it.

Here we test that alternative directly. We trained and archived 13 grading configurations on a pooled development set of 2,260 images, evaluated them on three external directions (3,662 APTOS images, 12,522 DDR images from 147 Chinese hospitals, and the reverse direction, APTOS to Messidor-2 and IDRiD), and asked, for every reported failure and every reported difference, whether it survives *matched calibration*: refitting the decision rule of every model by the same procedure before comparing them. Every training run and every hypothesis was pre-registered with its falsifying outcome before launch[15], and we report the interventions that failed with the same weight as those that worked.

We find that external failures of these models separate into two parts that call for different remedies. The ranking of images by severity transferred to both new populations, but the mapping from score to grade did not, and it moved in opposite

directions: the same models over-graded one population and under-graded another, so that a referral threshold fitted for 85% development sensitivity reached 99.33% on one and 63.38% on the other. Where the ranking survived, refitting the cut-points alone, with no retraining, recovered what had been lost: the mild grade on APTOS and much of the referable disease missed in the reverse direction. Where it did not, as for the foundation model on DDR, recalibration could not help. Contrary to two of our own pre-registered predictions, deployable rules fitted on development data did not move clinical metrics more than the choice of model, and the collapse of the mild grade was specific to one corpus. Quadratic weighted kappa (QWK), the field's standard summary, was nearly flat across operating points that differed by tens of points in referral sensitivity; models that differed only by random seed produced per-grade "effects" of the size that the literature attributes to architecture; and eight of nine pre-registered interventions on the representation were falsified. About 200 locally labelled images were enough to recalibrate safely.

Results

A pre-registered pipeline evaluated in three external directions

The development pool comprised 2,260 images (516 from IDRiD and 1,744 from Messidor-2) under a frozen 5-fold split; the external sets were APTOS 2019 (3,662 images) and DDR (12,522 images), and a separate reverse-direction model was trained on APTOS and tested on Messidor-2 and IDRiD (Fig. 1, Table 1). The selected model, an EfficientNet-B3 with two ordinal heads, was chosen on the development pool before any external number was examined. It reached a development QWK of 0.8954 at matched calibration and an external QWK of 0.9026 (95% CI 0.8941–0.9108) on APTOS and 0.7551 on DDR at its shipped decision rule. On identical images, folds and decoding, the full pipeline was worth +0.234 QWK over frozen ImageNet features with a linear ordinal head (QWK 0.6611). These numbers are competitive; the remainder of the paper asks what they do and do not measure.

Ranking transferred to new populations; the mapping from score to grade did not

Across the 10 pre-registered development models, the referral probability $P(\text{grade} \geq 2)$ ranked external images almost as well on both corpora: the area under the ROC curve was 0.914–0.973 on APTOS and 0.921–0.955 on DDR. Where those scores fell relative to the shipped cut-points did not transfer, and it moved in opposite directions on the two populations. Every model over-graded APTOS (mean predicted-minus-true grade +0.14 to +0.26) and under-graded DDR (−0.35 to −0.17). At the shipped rule (R1) this placed every model at a referable-DR sensitivity of 98.2–99.5% on APTOS but 58.9–70.3% on DDR (specificity 97.3–98.9%), and cut-points refitted on the external labels (R4, two-fold cross-fitted, a diagnostic) raised DDR sensitivity to 70.0–81.2% without retraining.

A referral threshold fitted on development predictions therefore did not hold its operating point (Fig. 3). Fitted for the 85% sensitivity of the superiority endpoint of

the first autonomous pivotal trial[5], the reference model referred 99.33% of referable patients on APTOS but 63.38% on DDR (pre-registered O8 held); fitted for 80%, it reached 98.12% and 50.96% (gaps +18.1 and −29.0 points). The same threshold sits at different points of the operating curve on different populations; a development-fitted operating point is not a property of the model.

The reverse direction showed the same dissociation. A model trained on APTOS with the same recipe (out-of-fold QWK 0.9134) under-graded Messidor-2 (mean predicted-minus-true grade −0.26; pre-registered O6 held) and, at its shipped rule, referred only 43.5% of referable eyes (specificity 99.2%), although it ranked them with an area under the curve of 0.890; refitted cut-points raised sensitivity to 77.9% (specificity 93.8%). The direction of miscalibration was set by the target population, not by the direction of transfer: the development models under-graded DDR just as the APTOS model under-graded Messidor-2.

The mild-grade collapse on APTOS is a decision-rule failure, but it is corpus-specific

On APTOS every development model lost the mild grade. At the shipped rule the models recalled 1.6–14.3% of the 370 mild-DR images (Fig. 2a), against 26.8–62.0% of mild images in the development pool at the same rule. Refitting only the four cut-points on half of the external labels and applying them to the other half (R4m) restored mild recall to 61.6–77.8% in every model, and the same held in all five seeds of the reference configuration (R1 5.4–6.2%, R4m 63.8–72.7%; pre-registered O3 held). Over-grading sent mild images across the moderate cut-point; the representation had separated them.

The collapse did not generalise beyond APTOS. On DDR, where the models under-graded, mild recall at the shipped rule was 20.5–45.2% and external cut-points raised it only to 51.6–62.1% (Fig. 2c). In the reverse direction, where a recent benchmark reported a mild-grade F1 of 0.000 for frozen encoders[9], our fine-tuned model recalled 34.4% of the 270 mild Messidor-2 images at the shipped rule, errors falling to no DR (175) rather than to moderate (2); the pre-registered predictions that it would collapse (O4) and that refitted cut-points would at least triple it (44.4%; O5) were both falsified. A collapsed grade is thus one possible consequence of a misplaced decision rule, not a general property of external DR grading.

Deployable rules did not repair the failures, and models differed more than rules

Rules fitted on development data did not transfer either. Cut-points chosen to maximise development QWK (R2) lowered external QWK for 9 of the 10 models on APTOS and left mild recall at 5.7–22.7%; a referral cut fixed to give 90% development sensitivity (R3) behaved likewise. We had pre-registered the prediction (O7) that the choice among these deployable rules would move mild recall and referable sensitivity more than the choice among models. It was falsified for both metrics on both corpora: on APTOS the median within-model range across R1–R3 was 5.7 points of mild recall against a between-model range of 12.7 at R1, and 0.9 against 1.3 points

of sensitivity; on DDR 21.0 against 24.8 and 7.1 against 11.4. Models trained on the same data thus differed materially at their shipped rules, and no rule fitted without external labels removed the population shift. The ranking of models was nonetheless rule-dependent: the Spearman correlation between rankings by external QWK at R1 and at the external-label rule R4 was 0.48 on APTOS and 0.55 on DDR, and the development ranking predicted the APTOS ranking with $\rho = 0.42$.

The grading metric cannot see the referral operating point

QWK, the field's summary statistic, was nearly invariant to the decision that determines who is referred. Holding the other cut-points at their QWK-optimal values and sweeping only the referral cut (Fig. 4), every operating point within 0.01 of the maximum QWK for the selected model on APTOS (0.9103) spanned referable sensitivity from 87.8% to 99.6% and specificity from 82.3% to 92.3%: between 6 and 182 of 1,487 referable patients missed, with QWK unable to tell them apart. On DDR the corresponding band spanned 71.3–86.3% sensitivity, between 773 and 1,615 of 5,626 referable patients missed. Optimising for the grade the metric under-weights carried its own cost: cut-points chosen to maximise macro-averaged recall recovered the mild grade but cut the reference model's APTOS referable sensitivity from 99.53% to 80.36%, because mild DR is not a referable grade. Macro-recall is therefore reported here as a diagnostic and never optimised.

The same pattern in a published model family

Because ORDER-DR[16] released image-level class probabilities for its models on both its APTOS validation split and the 1,744 Messidor-2 images, we could apply the same rules to an independently trained model family in the published direction, without any weights. Using the authors' own fusion and cut-points we reproduced their reported external result exactly (QWK 0.6423 ± 0.0364 over three seeds; 712.7 images predicted mild against 270 true, precision 0.193; Table 2). At argmax, the four released models recalled 2.0–5.2% of mild images and 32.8–36.3% of referable ones. With cut-points fitted on external labels (R4), all four converged to QWK 0.672–0.683: the 0.07 QWK advantage reported for the fused model over its best branch was a decision-rule effect. Refitting for macro-recall roughly doubled referable sensitivity (67.6–75.0%) and raised mild recall to 30.2–42.0%. Mild F1, however, stayed at or below 0.285: in this direction part of the mild failure is representational, and the decomposition, not either extreme, is the result.

Matched calibration changes model comparisons, and seeds set the noise floor

Comparisons between models were as sensitive to the decision rule as single-model metrics (Table 3). On APTOS the selected EfficientNet-B3 led the DenseNet121 reference by +0.0059 QWK [+0.0007, +0.0110] at the shipped rule; with both models' cut-points refitted the same way the difference became +0.0032 [−0.0020, +0.0083], and per-grade differences changed sign across operating points (moderate recall +1.4, +4.0 and −11.7 points under R1, R4 and R4m; severe −1.0, −10.3, +4.3). On the

development pool, linear-probe-then-fine-tune appeared to cost −5.54 accuracy points [-7.48, -3.54] at shipped cut-points and gained +1.95 [+0.18, +3.72] at matched ones. The rule is not a one-way filter: the development advantage of EfficientNet-B3 over its DenseNet121 control (+0.0202 shipped) survived matched calibration (+0.0207 [+0.0105, +0.0320]), and proliferative-DR recall favoured EfficientNet-B3 at every operating point.

Five seeds of the reference configuration set the scale of these effects. Development QWK at matched calibration varied with a standard deviation of 0.0043 across seeds (range 0.8646–0.8764), and APTOS QWK at the shipped rule with 0.0026. Between models that differed only by seed, 15 of 50 per-grade recall comparisons on APTOS were "significant" at the shipped rule (largest difference 8.2 points; pre-registered O2 held), and 29 at matched external cut-points; on DDR the counts were 17 and 21 of 30. An image-level bootstrap captures sampling of test images, not the variance of training, so per-grade differences of the size the literature attributes to architecture arise between identical architectures. A difference between configurations is treated here as beyond seed noise only if it exceeds $2\sqrt{2}$ seed standard deviations of matched development QWK (0.0123; Fig. 6).

About 200 local labels are enough to recalibrate safely

If cut-points must be refitted locally, the practical question is how many labelled local images that requires. Fitting cut-points on n randomly drawn APTOS images and evaluating on the rest (Fig. 5), 29.5% of draws at $n = 25$ and 16.0% at $n = 100$ ended worse than the shipped rule, although the mean gain was positive at every size. At $n = 200$ recalibration recovered +5.09 macro-recall points [+0.32, +7.28] with a 1.5% probability of harm, and at $n = 400$ the probability of harm was 0.0%. A recommendation judged by its mean alone would have endorsed a procedure that harms roughly one adopter in four at small n.

Interventions on the representation, and a supervision ceiling for macular oedema

Of 9 pre-registered interventions on the model itself, 8 failed their pre-specified criterion at matched calibration (Fig. 6; Supplementary Table S3). Only the EfficientNet-B3 backbone improved development QWK; a six-model ensemble gained on the development pool and +0.0025 [−0.0016, +0.0066] on APTOS. The retinal foundation model showed the clearest dissociation between representation and outcome: frozen, its features carried more DR signal than ImageNet features under an identical linear probe (0.7008 versus 0.6611, +0.0401 [+0.0122, +0.0692]); fine-tuned with linear-probe-then-fine-tune at its native 224 pixels, it fell below a DenseNet121 pretrained on EyePACS at the same resolution by −0.0414 [−0.0583, −0.0238] over 5 folds (pre-registered O9 held). Externally (a secondary analysis, outside the pre-registered model set) its two failures had different causes. On APTOS it ranked referable eyes as well as the convolutional models (area under the curve 0.947) and most of its QWK deficit was the decision rule: 0.703 at its shipped rule against 0.834 with refitted cut-points.

On DDR its ranking itself degraded (area under the curve 0.855, QWK 0.646 even with refitted cut-points), a representational failure that recalibration cannot repair.

No intervention moved the three-grade DME head. A crop centred on the fovea by a localiser validated before use (median error 0.196 disc diameters), which encodes the clinical definition of the grade directly, changed DME QWK by −0.0237 [−0.0566, +0.0094]. The constraint is supervision: only 51 of the 516 three-grade images (9.9%) carry the middle grade, a proportion that a purpose-built multimodal dataset reproduces (9.5%)[17], and a deployed commercial system reached a DME sensitivity of 26.5% (κ 0.38) on the same images on which its referable-DR κ was 0.81[8]. Nor can the grade be derived from published annotations: applying the definition (nearest hard exudate within one disc diameter of the fovea) to IDRiD's pixel-level exudate masks matched the expert grade in 96.3% of 54 images, against 92.6% for a constant predictor, and failed on two of the 4 images where the test was informative, each time on a single small exudate lying just inside the threshold (Supplementary Note 2).

Discussion

The central finding is that an external failure of an ordinal DR grading model has two separable components, and that published evaluations rarely separate them. The ranking of images by severity transferred well to both new populations; the mapping from score to grade did not, and its error had opposite signs on two populations, so that the same development threshold over-referred on one and missed a third of referable disease on the other. Where ranking survived, the loss was recoverable by moving cut-points alone: the mild grade on APTOS, which the literature reads as representational blindness to microaneurysms, and referral sensitivity in the reverse direction. Where ranking degraded, as for the foundation model on DDR, no recalibration helped. The analysis is also bounded by what it falsified. The choice among decision rules that can be deployed without external labels did not outweigh the choice of model, and the mild-grade collapse was specific to APTOS. A reader who sees an external failure of a grading model should therefore ask first whether the model still ranks the new population, then where its cut-points were set and on which population, and only then what it has learned.

This reframes several recent disagreements. Frozen-encoder benchmarks report that retinal foundation models fail on the mild grade externally[9], while the foundation model's own evaluation, by macro-averaged area under the curve, could not have detected such a failure[10]; our foundation-model results show the same dissociation from the other side, a better frozen representation that did not yield a better fine-tuned grader on 2,260 images. Calibration in the probabilistic sense is not the lever here: temperature scaling barely moves external calibration under shift[9], and in our data the referral probability was well ranked (external area under the curve 0.947 on APTOS) while the cut-points sat in the wrong place. Clinical evaluations show the same signature at the level of deployed systems. Commercial algorithms applied to the same eyes differed by more than 80 points in specificity[8], and seven systems on the same images spanned 35 points of sensitivity at similar negative predictive values[7]. Both are consistent with operating-point variation being reported as model variation,

and neither design could separate the two, because each system was evaluated only at its vendor's fixed operating point.

The reverse direction sharpens the point. Trained on APTOS, the same recipe did not lose the mild grade on Messidor-2, contrary to our own pre-registered prediction; it under-graded instead, and the cost fell on referral: at the shipped rule it referred 43.5% of referable eyes while ranking them with an area under the curve of 0.890, and refitted cut-points restored 77.9%. Which grade a population shift damages therefore depends on the population, and the development models under-graded DDR in the same way; in each case the damage lay in where the scores were cut rather than in how the images were ranked. A system validated as safe on its own population can miss more than half of referable disease on another without any change in its discriminative ability, and nothing in an agreement statistic would show it.

The metric invariance we observed explains why the problem persists. QWK charges little for a one-grade error, so a model that sends every mild case to moderate, or refers 99% rather than 88% of referable patients, scores almost the same. The same insensitivity has been observed in a deployed system, where agreement with the reference standard moved from κ=0.65 to 0.72 while sensitivity and specificity each moved by about 31 points between phases, although that change also involved retraining and a new cohort[8]. Metric choice is a known source of misleading validation in image analysis[18,19]; for screening, the implication is concrete: an agreement statistic cannot stand in for the referral operating point, and the two must be reported together.

The practical recommendations follow directly and cost no additional training. Model comparisons should be reported at matched calibration, with the fitting procedure named, because a per-grade verdict can depend on it (Supplementary Note 3). External evaluations should report the operating curve and the transfer gap of any development-fitted threshold rather than a single sensitivity, since the same threshold landed +18.1 points from its target on one population and −29.0 on another. Deployment should include local recalibration of the cut-points on about 200 labelled images, a procedure whose probability of harm, not only its mean benefit, we report, in line with long-standing advice on updating prediction models for new settings[13,14] and with the sample sizes recommended for external validation[20]. The recalibration need not change the network: it is a re-estimation of four numbers.

The limits of the argument matter as much as its reach. First, decision rules do not explain everything. In the APTOS-to-Messidor-2 direction, refitting the cut-points of a published model raised mild recall several-fold but left mild F1 below 0.3, and in our own model trained on APTOS external cut-points raised mild recall only to 44.4% (pre-registered O5 falsified), so part of the mild-grade failure in that direction is representational; and the internal mild-grade errors of our own models were only partly recoverable. The claim is that the decision rule must be separated before the representation is blamed, not that representation never matters. Second, the cut-points that recover the mild grade were fitted on external labels; they bound what local recalibration can achieve and are not themselves external validation, which is why the deployable rules and the sample-size analysis are reported separately. Third, all data are public and retrospective. APTOS labels come from a single grader, so part of any internal-to-external difference is label noise; none of the development or external

corpora publishes patient identifiers, so resampling is by image and fellow-eye pairs may be correlated across folds; EyePACS, used for pretraining, also contributes to the retinal foundation model's pretraining corpus[10]; and the checkpoint epoch was selected on the held-out fold, which makes development estimates slightly optimistic (median +0.0025 QWK) without affecting external ones. Fourth, the macular-oedema head has no external validation: the one candidate corpus overlapped the development pool by 88%, and the three-grade label could not be derived from published annotations, so our DME conclusions rest on 516 images and are stated as a supervision ceiling rather than a performance claim. Finally, the reverse direction and the seed replicates use a single architecture, and the published-model reanalysis uses one model family; the decision-rule effect should be tested in prospective screening data, where cut-points can be fitted on an initial local sample and then frozen.

In summary, external failures of DR grading models are routinely attributed to learned representations, but a substantial part of them, and of the apparent differences between models, lies in the decision rule that converts scores into grades, and that part can be diagnosed and repaired without retraining. Reporting threshold-free discrimination beside matched-calibration comparisons, full operating curves and the transfer gap of any threshold, and recalibrating cut-points locally before deployment, would separate the two components and make published results comparable with one another and with the clinical decision they are meant to support.

Methods

Study design and pre-registration

This is a retrospective modelling study on public, de-identified fundus photographs. It was conducted under an evaluation protocol frozen before the first training run (unit of independence, held-out sets, model selection on the development pool only, metrics and uncertainty), and every experiment was pre-registered: its hypothesis, control and falsifying outcome were committed to the version-controlled repository before the run was launched, and the commit history is the record[15]. The nine interventions on the model and the nine outcomes O1–O9 of the journal experiments (seed replicates, the reverse direction, the second external corpus and the foundation-model completion) were registered in this way; a training run cannot be launched by our kernel builder without a recorded hypothesis. Reporting follows TRIPOD+AI[21] and CLAIM[22] where applicable (Supplementary Table S1).

Datasets

The *development pool* combined IDRiD[23,24] (516 images; 5-grade International Clinical DR scale[25] and a 3-grade risk-of-DME label defined by the distance of the nearest hard exudate to the macula) and Messidor-2[26,27] (1,744 gradable images with adjudicated DR grades and a binary referable-DME label[28]), 2,260 images in total (Table 1). IDRiD's official train/test partition was pooled, because a 103-image test split cannot resolve the effects studied here (half-width of a 95% interval on accuracy ≈6 points). The *external* DR sets were APTOS 2019 (3,662 images from Aravind Eye Hospital,

India; single-grader labels) and DDR[29] (12,522 gradable images from 147 hospitals in 23 Chinese provinces, graded by seven graders; images distributed at 512 pixels). Neither was used for training, model selection or threshold fitting, and both were held out from the first run. The EyePACS 2015 training set (35,126 images, single-grader labels) was used only to pretrain the DR head. In the reverse-direction experiment APTOS was the training corpus and Messidor-2 and IDRiD the external sets. Per-image identity checks found no image of any external set in any training set. No patient identifiers are published for IDRiD, Messidor-2 or APTOS; the unit of resampling is therefore the image, and near-duplicate images were merged into one group by a 256-bit difference hash calibrated on fundus images (distinct pairs $\geq$28 bits apart, re-encodings of one image $\leq$8; threshold 16).

Labels

DR labels are the 5-grade international scale in all corpora (0 no DR, 1 mild, 2 moderate, 3 severe non-proliferative, 4 proliferative); referable DR is grade $\geq$2. DME is graded 0–2 in IDRiD (none; exudates more than one disc diameter from the macula; within one disc diameter). Messidor-2's binary referable-DME label is, by its definition, exactly IDRiD's grade 2; a non-referable Messidor-2 image is grade 0 *or* 1. In threshold form this partial label supervises the second DME threshold and leaves the first unsupervised for that image, so no label is imputed. The 3-grade DME metrics are computed on IDRiD only (516 images, 51 of them in the middle grade), always beside the majority-class floor (47.1%), and without gating on DR.

Model and training

All models share one design: a convolutional (or transformer) backbone, global average pooling, and two ordinal threshold heads (hidden width 256, dropout 0.4) that output $K-1$ logits for $P(y > k)$, four for DR and two for DME[30]. The loss is $0.6\,\mathcal{L}_{\mathrm{DR}} + 0.4\,\mathcal{L}_{\mathrm{DME}}$, each a masked binary cross-entropy over the thresholds with label smoothing 0.05. Images were cropped to the retinal disc, cached at a long side of 560 pixels and resized to the training resolution (448 pixels unless stated), with unrestricted rotation, scaling (0.88–1.12), translation ($\pm$4%), horizontal and vertical flips and mild photometric jitter; a sampler balanced the DR grades. Training used AdamW (backbone learning rate 10^{-4}, heads 10^{-3}, weight decay 10^{-4}), a 5% linear warm-up followed by cosine decay to 2% of the peak, gradient clipping at 1.0, mixed precision, and an exponential moving average of the weights (decay 0.999) for evaluation. The reference configuration (E08) is DenseNet121[31] pretrained on ImageNet, then on EyePACS for 5 epochs, then fine-tuned for 40 epochs with batch 16. The selected model (E11FULL) replaces the backbone with EfficientNet-B3[32] (30 epochs). Every other configuration changes exactly one factor relative to its control (Supplementary Table S2): resolution (224, 560 and native 640 pixels), a macula-centred crop for the DME head driven by a fovea localiser validated before use (median error 0.196 disc diameters), linear-probe-then-fine-tune[33], CORAL and CORN ordinal heads[30,34], ConvNeXt-tiny[35], and the RETFound colour-fundus ViT-L/16 encoder[10] (224 pixels, linear probe for 5 epochs then fine-tuning at 10^{-5}). At the end of each fold the

epoch with the highest $0.6\,\mathrm{QWK}_{\mathrm{DR}} + 0.4\,\mathrm{QWK}_{\mathrm{DME}}$ on the held-out fold was kept, and its predictions were averaged over the four dihedral flips (test-time augmentation). External predictions average the logits of the five fold models.

Cross-validation and model selection

All development results are out-of-fold over the full pool: a 5-fold split, stratified by corpus and DR grade and grouped by near-duplicate cluster, was built once, fingerprinted and committed before the first run, and every model loads it from that file. Configurations were ranked by development QWK at matched calibration (below) and the selected model was declared in the repository before any external number was examined; no external result was used to choose a model, a threshold or a hyperparameter. Because the checkpoint epoch was chosen on the held-out fold, development estimates carry a small selection optimism. We estimated it by re-selecting each fold's epoch from the *other* folds' validation curves: across nine 5-fold runs the median optimism was +0.0025 DR QWK (range −0.0008 to +0.0159). External estimates are not affected.

Decision rules and matched calibration

For an ordinal model with threshold probabilities $p_k = P(y > k)$ the expected grade is $s = \sum_k p_k$. We compare five decision rules, fixed before the journal experiments. **R1** is the shipped rule: grade = the number of leading thresholds with $p_k > 0.5$ (made rank-consistent with a cumulative minimum). **R2** places four cut-points on s to maximise QWK on the development out-of-fold predictions. **R3** is R2 with the grade-1/2 cut moved to the most specific value that gives a development referable sensitivity of at least 0.90. R1–R3 are deployable: none uses external labels. **R4** and **R4m** fit the cut-points on the external labels themselves, maximising QWK or macro-averaged recall, by two-fold cross-fitting (each half's cut-points are fitted on the other half); they are diagnostic upper bounds on what recalibration with local labels can recover and are never reported as performance. Cut-points are found by coordinate ascent (eight rounds; 60 candidates per cut on development data and 80 on external data, reproducing the archived analyses exactly; for the reference model the grid alone moves matched QWK by 0.0033); full definitions are given in Supplementary Note 1.

Matched calibration is the comparison rule applied throughout: no difference between two models is attributed to representation unless it survives refitting both models' cut-points by the same cross-fitted procedure (on the development pool, fold f's cut-points come from the other four folds). The rule is only evidence if it can return either answer; its record in this study is reported in Table 3. Because a per-grade verdict can depend on how the cut-points are cross-fitted (Supplementary Note 3), every matched claim names its procedure.

Metrics and statistical analysis

The primary grading metric is QWK[36]; the operationally primary pair is referable-DR sensitivity and specificity, never quoted alone; per-grade recall is reported beside every

headline, and macro-averaged recall is used as a diagnostic, never as an objective (optimising it costs referral sensitivity; see Results). We also report Cohen's unweighted κ, accuracy and macro-F1, and, for the referral probability $P(\text{grade} \geq 2) = p_1$, the area under the ROC curve, the Brier score and the expected calibration error (ECE, ten equal-width bins) together, since ECE alone can reward an uninformative predictor[9]. Intervals are 95% percentile intervals from 2,000 bootstrap resamples of groups; differences between models are paired bootstraps over the same groups, and a difference whose interval contains zero is reported as indistinguishable, not ranked. No correction for multiple comparisons is applied; per-grade claims are therefore treated as credible only when they replicate across conditions.

Recalibration sample size

To quantify how many locally labelled images recalibration needs, we drew $n \in \{25, 50, 100, 200, 400\}$ external images at random (200 draws per size), fitted cut-points on those n alone (maximising macro-recall), and evaluated on the remaining images, so that every estimate is out of sample with respect to its cut-points, as in deployment. The probability of harm is the fraction of draws in which macro-recall fell below that of the shipped rule on the same held-out images.

Seed replicates, reverse direction and second external corpus

Seeds 1–4 of the reference configuration were trained with every other setting identical (E08 is seed 0; consumption manifests confirm the same files were read), and each was scored on APTOS and DDR. The reverse-direction model used the reference recipe trained on APTOS with a 5-fold split built by the same code before training (near-duplicates merged), and was scored on Messidor-2 and IDRiD (Supplementary Notes 5 and 6). The archived development models were re-scored on DDR (and those without archived APTOS predictions on APTOS) from their stored fold weights, with the same inference recipe. Because public fundus corpora draw on overlapping sources, every image of the development pool, the external sets and EyePACS was compared by a 256-bit difference hash after the training crop, and near-duplicate pairs across corpora were counted (Supplementary Note 7).

Provenance controls

Each run records, from inside the process that read the disk, a consumption manifest (number of images per corpus and a hash over the resolved paths); runs whose manifests differ cannot be combined or compared without the difference being printed into the output. Every number in this article is generated from archived artefacts into a ledger, and the manuscript is compiled from that ledger, so no number in the text is typed by hand.

Use of AI assistance

An AI system (Claude, Anthropic) assisted with analysis code, the execution of the pre-registered journal experiments and drafting of the manuscript under the direction

of the authors, who verified the analyses, the numbers against their generated sources, and every cited claim against its primary source, and take full responsibility for the content.

Data availability

All images are from public datasets: IDRiD (IEEE DataPort), Messidor-2 (ADCIS, with adjudicated grades from Krause *et al.*[28]), APTOS 2019 and EyePACS 2015 (Kaggle) and DDR[29]. Per-image out-of-fold and external predictions for every model, the frozen split, the pre-registration records and every generated table are available in the project repository (https://github.com/Ahmadrezanourozii/Automatic-detection-of-diabetic-retinopathy-and-grading-of-diabetic-macular-edema-using-CNN).

Code availability

All code (training, evaluation, decision-rule analyses, figure and number generation) is available in the same repository; each run is reproducible from the commit hash recorded in its log.

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Author contributions. [To be completed by the authors, referring to each author by initials, e.g. A.C. conceived the study, implemented the pipeline, ran the experiments and wrote the manuscript; ...]

Competing interests. The authors declare no competing interests. [Confirm for each author.]

Ethics. This study used only publicly available, de-identified images released for research; no new data were collected from human participants and no ethical approval was required. [Confirm against institutional policy.]

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Table 1 Cohorts. No external image was used for training, model selection or threshold fitting. DR, diabetic retinopathy (international 5-grade scale); DME, diabetic macular oedema.

Role	Corpus	Origin	Images	Labels	Grading
Development pool	IDRiD	India	516	DR 0–4; DME 0–2	3 graders (published grading)
	Messidor-2	France	1,744	DR 0–4; referable DME	adjudicated (3 specialists)
Pretraining only	EyePACS 2015	USA	35,126	DR 0–4	single grader
External (forward)	APTOS 2019	India	3,662	DR 0–4	single grader
	DDR	China (147 hospitals)	12,522	DR 0–4	7 graders
Reverse direction	APTOS → Messidor-2, IDRiD		3,662 → 2,260	DR 0–4	as above

Table 2 Decision-rule reanalysis of a published model family (ORDER-DR, APTOS→Messidor-2). Mean ± s.d. over the authors' three seeds, computed from their released image-level probabilities on 1,744 Messidor-2 images (270 mild). For ORDER-DR the R2 row uses the authors' own fusion and cut-points and reproduces their reported result. R4/R4m use external labels (two-fold cross-fitting) and are diagnostic bounds, not performance claims.

Model	Rule	QWK	Mild rec. (%)	Mild F1	Sens. (%)	Spec. (%)
CB-384	R1 argmax	0.500 ± 0.018	2.0	0.027	32.8	99.5
	R2 val.-fitted	0.544 ± 0.029	2.5	0.035	38.2	99.0
	R4 ext. QWK	0.672 ± 0.030	27.8	0.231	55.1	95.4
	R4m ext. recall	0.614 ± 0.060	38.4	0.255	72.6	85.2
LORS-384	R1 argmax	0.531 ± 0.014	3.5	0.049	36.3	98.9
	R2 val.-fitted	0.570 ± 0.023	2.7	0.041	42.8	98.7
	R4 ext. QWK	0.683 ± 0.045	30.5	0.245	55.9	96.1
	R4m ext. recall	0.677 ± 0.046	36.5	0.285	67.6	91.3
Swin-T	R1 argmax	0.552 ± 0.045	5.2	0.062	35.4	96.9
	R2 val.-fitted	0.564 ± 0.026	3.3	0.045	43.2	96.2
	R4 ext. QWK	0.674 ± 0.022	13.0	0.147	64.8	93.8
	R4m ext. recall	0.629 ± 0.041	30.2	0.233	75.0	88.9
ORDER-DR	R1 argmax	0.509 ± 0.017	2.7	0.040	34.1	99.2
	R2 val.-fitted	0.642 ± 0.036	50.5	0.277	39.8	99.1
	R4 ext. QWK	0.676 ± 0.025	24.2	0.212	56.7	95.4
	R4m ext. recall	0.638 ± 0.049	42.0	0.284	70.0	88.8

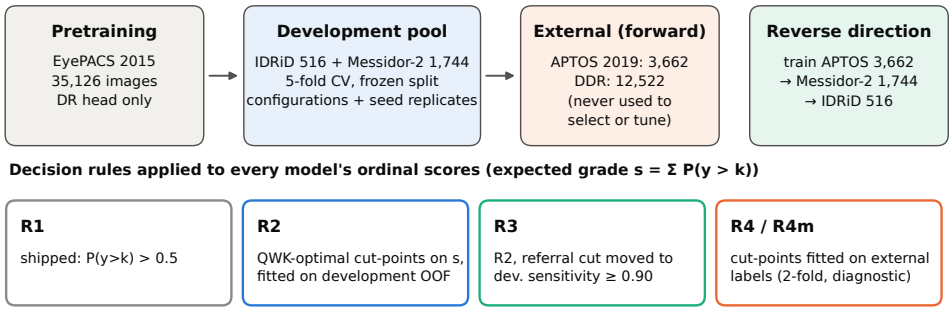


Fig. 1 Study design. Models were pretrained on EyePACS, developed on a pooled IDRiD and Messidor-2 set under a frozen 5-fold split, and evaluated without any further tuning on APTOS 2019 and DDR; a separate model trained on APTOS with the same recipe was evaluated on Messidor-2 and IDRiD (reverse direction). Every model's ordinal outputs were decoded under the same five decision rules: R1, the shipped default; R2 and R3, deployable rules fitted on development predictions; and R4/R4m, diagnostic rules fitted on the external labels by two-fold cross-fitting, which bound what local recalibration can recover and are never reported as performance.

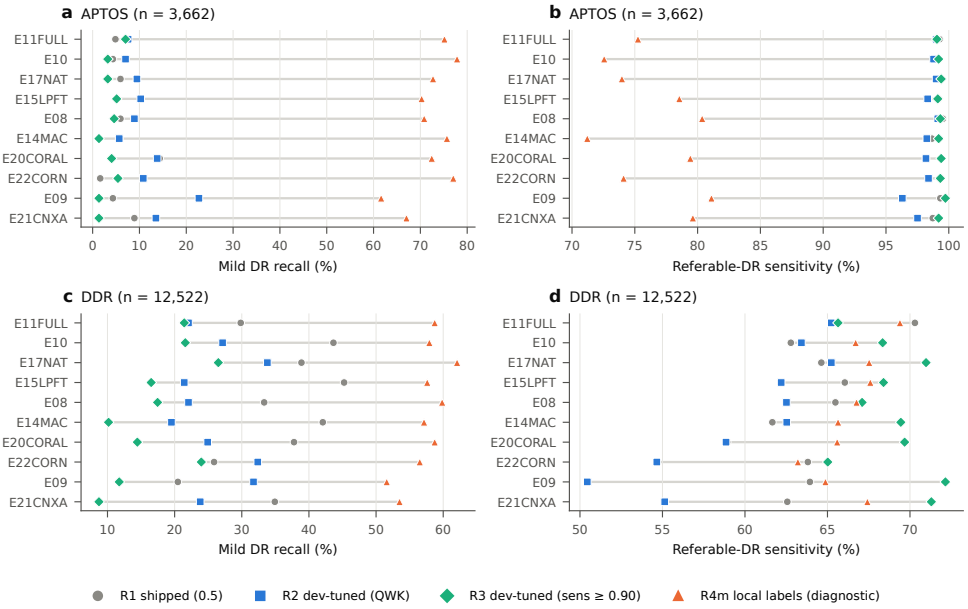


Fig. 2 External mild-grade recall and referral sensitivity of every pre-registered model under every decision rule. Each row is one development model; markers show the same model's external metric under four decision rules. **a**,**b**, APTOS 2019; **c**,**d**, DDR. On APTOS, mild recall under the shipped rule (R1) is low in every model and is restored by cut-points refitted on half of the external labels (R4m, a diagnostic that involves no retraining); development-fitted rules (R2, R3) do not restore it. On DDR the mild grade is not collapsed and external cut-points recover less. Models differ at a fixed rule by more than the deployable rules R1–R3 differ within a model (pre-registered outcome O7, falsified). Referral sensitivity under R4m falls because mild DR is not a referable grade.

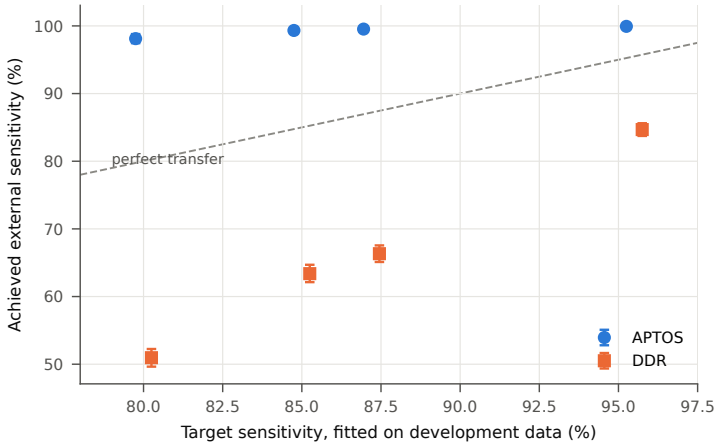


Fig. 3 Thresholds fitted on development data do not hold their operating point on new populations. A referral threshold on $P(\text{grade} \geq 2)$ was fitted on the development out-of-fold predictions of the reference model to reach each target sensitivity (x-axis), then applied unchanged to APTOS and DDR. Points show the achieved external sensitivity with 95% bootstrap intervals; the diagonal is perfect transfer. The same thresholds over-refer on APTOS, where the models over-grade, and under-refer on DDR, where they under-grade. Targets: 80%; 85% (the pre-specified superiority endpoint of the first autonomous pivotal trial); 87.2% (that trial's achieved sensitivity); 95.5% (a reported pivotal-trial sensitivity).

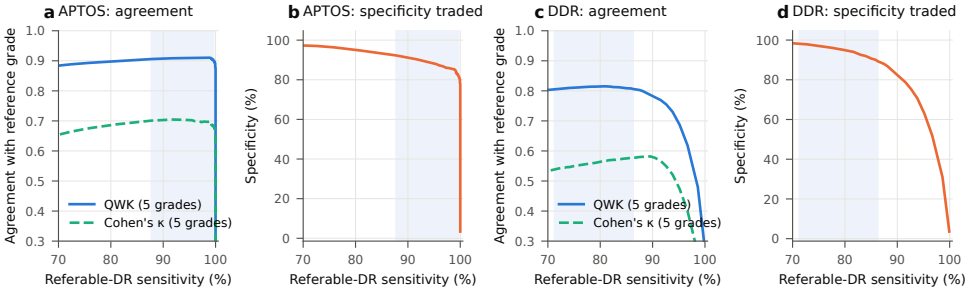


Fig. 4 Agreement statistics cannot see the referral operating point, and development-fitted thresholds do not transfer. a,**c**, QWK and Cohen's κ of the selected model as the referral cut-point is swept with the other cut-points held at their QWK-optimal values, plotted against the referable-DR sensitivity it produces, on APTOS (**a**) and DDR (**c**). The shaded band contains every operating point whose QWK is within 0.01 of the maximum. **b**,**d**, the specificity that the same sweep trades for sensitivity. Inside the band the referral decision changes by tens of points while the grading statistic does not.

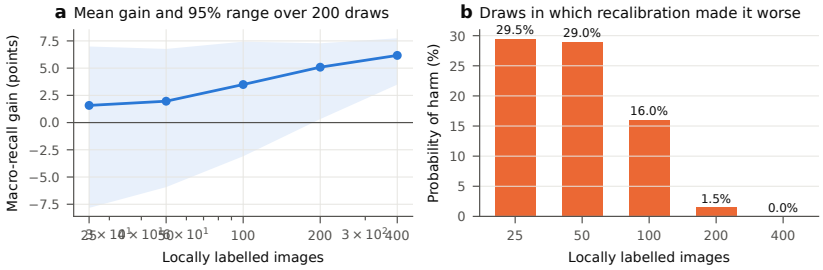


Fig. 5 How many locally labelled images recalibration needs. Cut-points were fitted (maximising macro-averaged recall) on n APTOS images drawn at random and evaluated on the remaining images, 200 draws per size. **a**, mean gain in macro-averaged recall over the shipped rule with the 2.5th–97.5th percentile range of draws. **b**, probability that a draw ends worse than the shipped rule. The mean gain is positive at every size, but below 100 images roughly one recalibration in four does harm.

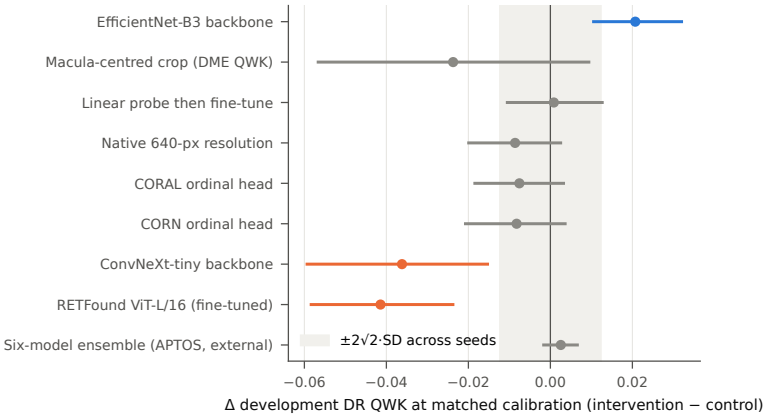


Fig. 6 Pre-registered interventions on the model. Difference in development DR QWK (intervention minus control) at matched calibration with 95% paired-bootstrap intervals; the macula-centred crop was registered on the DME head (DME QWK shown) and the ensemble on the external set (APTOS QWK shown). The grey band is $\pm 2\sqrt{2}$ standard deviations of the matched development QWK across five seeds of the reference configuration, the pre-registered yardstick for a difference beyond seed noise. Blue, pre-registered improvement confirmed; orange, interval excludes zero in the unfavourable direction; grey, indistinguishable.

Table 3 Matched calibration changes model comparisons. Differences (first minus second) with 95% paired-bootstrap intervals at the shipped rule and with both models' cut-points refitted by the same procedure (external rows: R4, two-fold cross-fitted on APTOS labels; development rows: fold-wise cross-fitted). *Interval excludes zero. †Under random-half cross-fitting the matched difference is significant; under fold-wise cross-fitting it is not (Supplementary Note 3).

Comparison	Shipped rule	Matched calibration	Verdict
EfficientNet-B3 vs DenseNet121, APTOS QWK	+0.0059 [+0.0007, +0.0110]	+0.0032 [−0.0020, +0.0083]	advantage dissolves
recall, moderate (points)	+1.4	+4.0*	significance changes
recall, severe (points)	−1.0	−10.3*	significance changes
recall, proliferative (points)	+8.1*	+6.8*	persists
LP-FT vs fine-tuning, development accuracy (points)	−5.54 [−7.48, −3.54]	+1.95 [+0.18, +3.72]	reverses
EfficientNet-B3 vs DenseNet121, development QWK	+0.0202 [+0.0071, +0.0329]	+0.0207 [+0.0105, +0.0320]	survives
Image source (E12), moderate recall (points)†	−6.9 [−12.1, −2.1]	+0.4 [−5.7, +6.1]	procedure-dependent